RTL-to-GDSII Implementation of Digital Building Blocks for Fractional-N DPLL

By:

Karim Mohamed Abomera

July 16, 2026

Chapter 1

Introduction

1.1Overview

With the growing demand for high-performance and energy-efficient wireless systems, digital-intensive architectures have become essential in modern frequency synthesis. Fractional-N Phase-Locked Loops (PLLs) rely heavily on advanced digital blocks such as Digital Loop Filters (DLF), Sigma-Delta Modulators (ΣΔ), and correlator-based fractional division techniques to achieve high resolution and low noise performance.

This project focuses on the design and full ASIC implementation of these key digital building blocks, starting from RTL description and progressing through the complete physical design flow until GDSII generation

1.2 Block diagram

First, we will take an overview of the overall architecture of the proposed system, as illustrated in Fig.1.

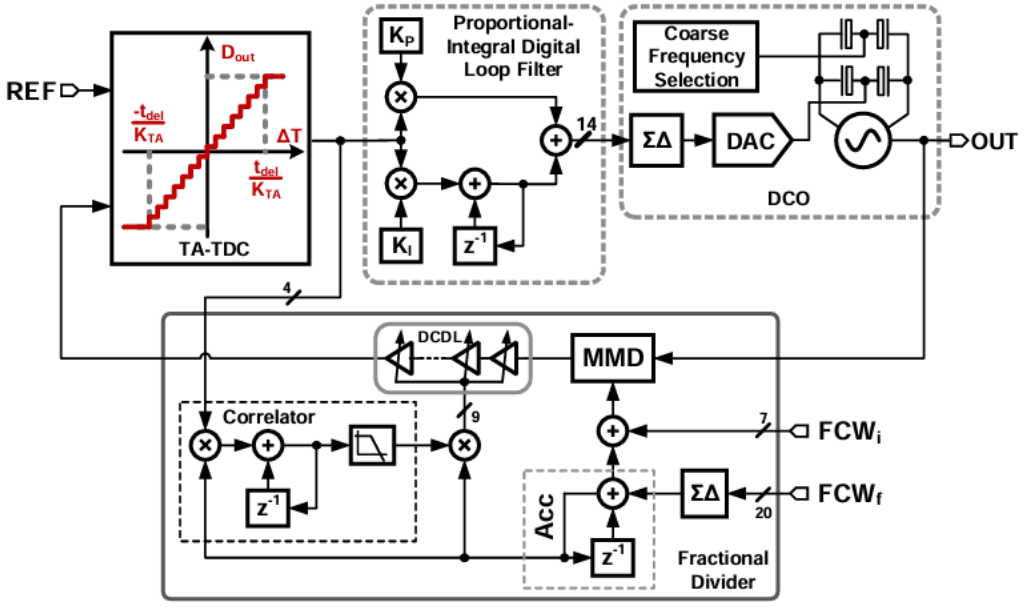


Fig.1 : Detailed block diagram of the proposed Digital FNPLL

1.3 How the Loop Locks

The system operates by converting the phase difference between the reference and feedback signals into a digital value using a Time-to-Digital Converter (TDC). This error is filtered by the Digital Loop Filter (DLF) and generates the Digital Control Word (DCW) that is delivered to the Digital Delta Sigma Modulator (DDSM). The **DDSM** converts a 14-bit fractional input to a 5-bit integer output that is delivered to the Digitally Controlled Oscillator (DCO)**.** The **DCO** adjusts frequency based on the **DCW**, and then the feedback loop locks the output to the desired frequency, while the correlator reduces spurious tones and enhances overall system performance.

Chapter 2

Digital Building Blocks Design and Implementation

2.1 Introduction

Following the overview of the ASIC design methodology presented in Chapter 1, this chapter focuses on the design and implementation of the digital building blocks developed for the proposed Fractional-N Digital Phase-Locked Loop (DPLL). Each block was designed using Verilog, functionally verified through simulation, and subsequently integrated into the overall digital system.

The objective of this chapter is to present the architecture, operating principle, hardware implementation, and functional verification of each digital block individually before discussing their integration into larger subsystems. Designing and validating each block independently simplifies debugging, facilitates functional verification, and ensures that every module satisfies its intended specification prior to system-level integration.

The implementation methodology and verification results presented in this chapter establish the foundation for the ASIC implementation discussed in the following chapter.

2.2 Digital Loop Filter (DLF)

2.2.1 Role of the Digital Loop Filter

The Digital Loop Filter (DLF) is one of the fundamental building blocks of a Digital Phase-Locked Loop (DPLL). Its primary function is to process the digital phase error generated by the PFD and produce a stable control signal that drives the Digitally Controlled Oscillator (DCO). By filtering the instantaneous phase error and accumulating long-term frequency deviations, the DLF enables the DPLL to achieve accurate phase and frequency synchronization while maintaining loop stability.

2.2.2 Why Digital?

Unlike conventional analog PLLs that rely on large RC loop filter and loop filter is typically implemented using passive or active analog circuits, the DLF performs all filtering operations digitally. This digital implementation offers several advantages, including improved scalability across CMOS technology nodes, immunity to process, voltage, and temperature (PVT) variations, ease of parameter tuning, and seamless integration with other digital components of the system.

2.2.3 operating principle

In the proposed Fractional-N DPLL, the DLF receives a 4-bit digital phase error generated by the Time-to-Digital Converter (TDC). This error represents the phase difference between the reference clock and the feedback clock. The DLF processes the error using a Proportional–Integral (PI) control algorithm to produce a 15-bit digital control word that is delivered to the DDSM, then determines the operating frequency of the Digitally Controlled Oscillator (DCO).

Fig.2.1 illustrates the implemented DLF adopts a Type-II Proportional-Integral (PI) controller, which consists of two parallel paths:

- **Proportional (KP) path**: Provides an immediate response to the current phase error, improving the transient response and reducing the settling time of the loop.
- **Integral (KI) path**: Accumulates the phase error over time, eliminating the steady-state error and improving long-term frequency accuracy.

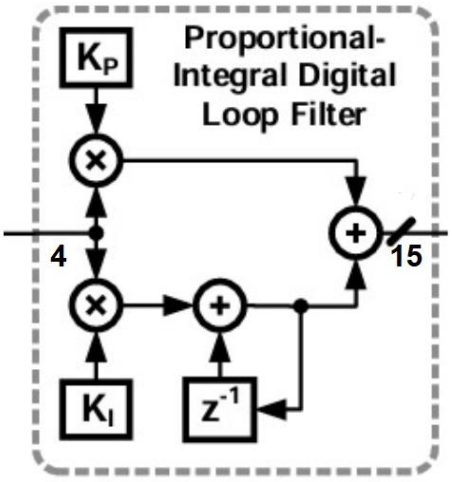


Fig.2.1: Block diagram of the implemented DLF architecture.

The combination of these two control actions enables the loop to achieve both fast dynamic performance and zero steady-state error under normal operating conditions.

Consequently, the performance of the DLF has a direct impact on the overall stability, convergence speed, and frequency accuracy of the proposed DPLL.

2.2.4 Simulation Results

The outputs of both paths are summed to generate the final digital control word. The input phase error is multiplied by programmable proportional and integral coefficients, making the design suitable for ASIC implementation while maintaining high numerical accuracy.

To ensure reliable operation, the integral accumulator incorporates overflow and underflow protection. Instead of allowing arithmetic wrap-around, the accumulator employs clipping logic. Whenever the accumulated value exceeds the supported numerical range, the output is clipped to either the maximum positive or maximum negative value, which could otherwise cause instability or significant frequency disturbances within the DPLL.

Fig.2.2 illustrates the simulation results confirmed the correct generation of the control word as well as the proper assertion of the overflow and underflow flags whenever the accumulator reached its operating limits.

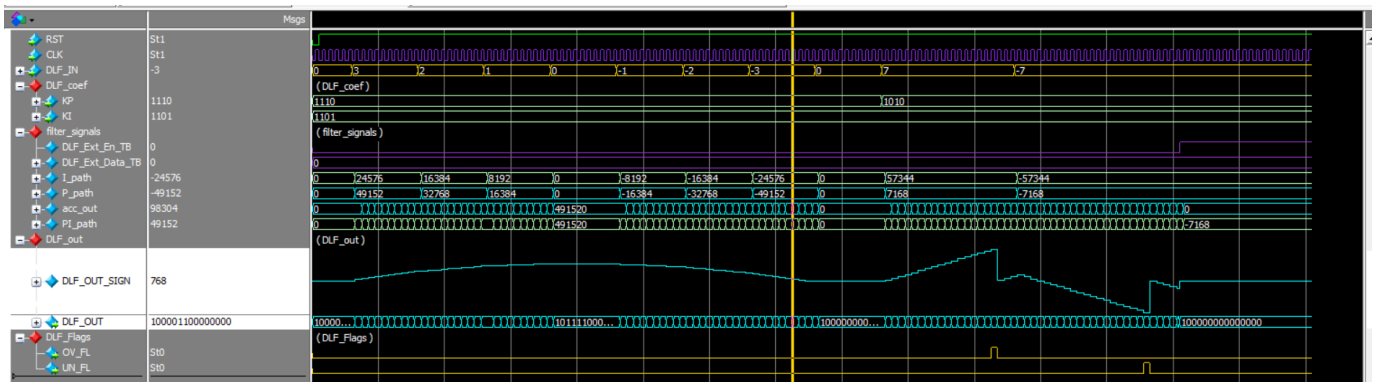


Fig. 2.2: Simulation waveform showing the DLF operation.

The signals **DLF_Ext_En** and **DLF_Ext_Data** are implemented to facilitate the testing of the Digitally Controlled Oscillator (DCO) frequency tuning range and resolution. By enabling external control, these signals bypass the digital loop filter. This technique provides a way to examine the DCO frequency tuning and its range and resolution as a standalone circuit

2.3 First-Order Digital Delta Sigma Modulator (DDSM)

2.3.1 Introduction

Fractional-N frequency synthesizers achieve fine frequency resolution by allowing the feedback division ratio to vary between adjacent integer values. Instead of using a fixed integer division ratio, the divider alternates between neighboring integer values such that the average division ratio equals the desired fractional value. Although this technique significantly improves frequency resolution, it introduces quantization error that can generate undesirable spurious tones and degrade the spectral purity of the synthesized output.

2.3.2 operating principle

Digital Delta Sigma Modulator (DDSM) is a well-established digital signal processing technique used to mitigate this problem. Rather than eliminating quantization error, a DSM redistributes its spectral energy by shaping the quantization noise toward higher frequencies. Since the DPLL behaves as a low-pass filter with respect to divider noise, a large portion of this high-frequency noise is naturally attenuated by the loop, thereby reducing the in-band phase noise and improving the overall spectral performance.

Among the various Sigma-Delta architectures, the first-order Sigma-Delta Modulator provides the simplest implementation while effectively demonstrating the principle of quantization noise shaping. Owing to its low hardware complexity, small silicon area, and ease of implementation, it is widely employed in digital frequency synthesizers and serves as the fundamental building block for higher-order architectures such as the MASH (Multi-stage Noise Shaping) modulator.

The modulator receives a 20-bit fractional control word generated by the digital processing stage and produces a sequence of integer control values whose average corresponds to the desired fractional division ratio. This sequence is subsequently used to control the programmable divider, enabling fractional frequency synthesis while shaping the quantization noise away from the loop bandwidth.

Fig.2.3 illustrates the block diagram consists of an accumulator and a quantizer that separates the integer and fractional portions of the accumulated value.

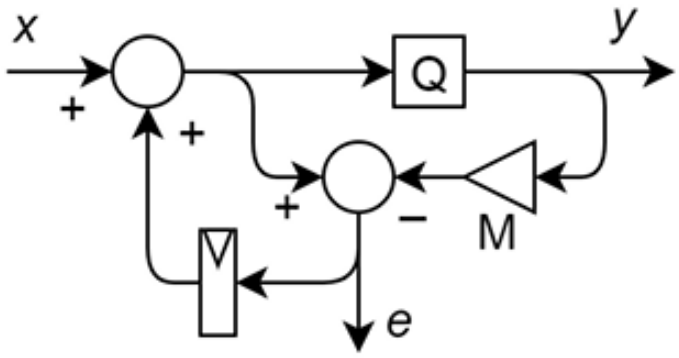


Fig2.3: Block Diagram of first-order DDSM.

Fig.2.4 illustrates the simulation results confirmed the correct operation of both accumulator stage, quantizer process and the expected first-order noise-shaping behavior.

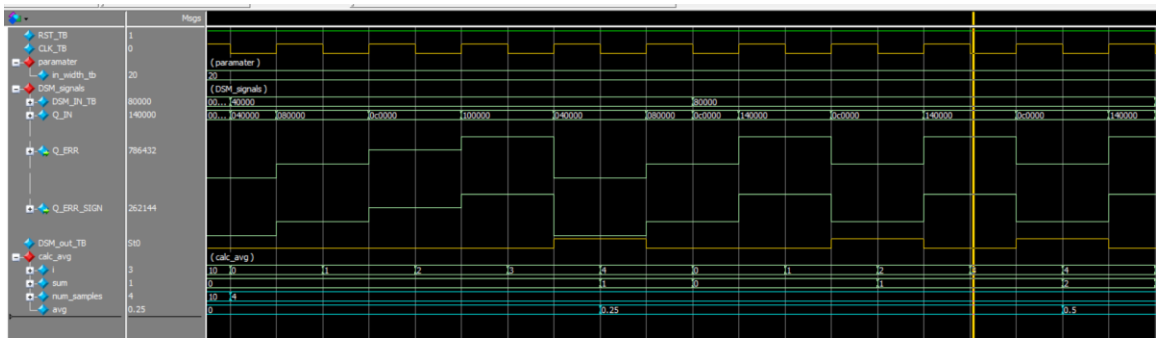


Fig2.4: Simulation waveform showing the output

2.4 Second-Order Sigma-Delta Modulator (MASH 1-1)

2.4.1 Introduction

As the frequency resolution of Fractional-N Digital Phase-Locked Loops (DPLLs) increases, the impact of quantization noise becomes more significant. Although a first-order Sigma-Delta Modulator effectively shifts a portion of the quantization noise toward higher frequencies, its noise shaping capability is often insufficient for applications requiring low in-band phase noise and improved spectral purity.

The first-order DSM output sequence may exhibit periodic patterns for certain fractional input values. These repetitive sequences can produce idle tones and fractional spurs that degrade the spectral purity of the synthesized output.

To alleviate this limitation, the proposed design employs a MASH 1-1 Sigma-Delta Modulator. By cascading two first-order stages and applying digital error cancellation, the MASH architecture significantly reduces the periodicity of the output sequence. Consequently, the generated bit stream becomes less repetitive, effectively dispersing the quantization error over time rather than allowing it to concentrate at discrete spectral components.

The Multi-stage Noise Shaping (MASH) structure has become one of the most widely adopted solutions due to its excellent stability, modular implementation, effective high-order noise shaping and less in-band quantization noise without significantly increasing hardware complexity.

2.4.2 operating principle

Fig.2.5 illustrate a MASH 1-1 modulator which realized by cascading two first-order Sigma-Delta Modulators and combining their outputs through a digital noise cancellation network according to

$$y[n]=y_1[n]+y_2[n]-y_2[n-1] \quad (1)$$

Unlike a conventional second-order Sigma-Delta Modulator, which employs a single second-order feedback loop, the MASH architecture consists of multiple stable first-order stages. This approach preserves the unconditional stability of first-order modulators while achieving second-order quantization noise shaping.

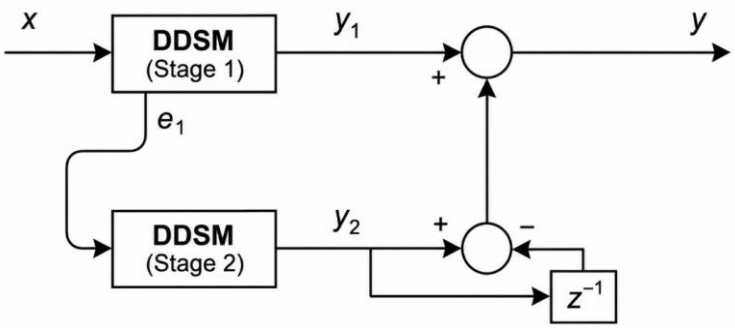


Fig.2.5: Block Diagram of the implemented MASH 1-1 DSM

In the proposed Fractional-N DPLL, a MASH 1-1 Sigma-Delta Modulator is employed to process the fractional component of the Digital Control Word (DCW) generated by the Digital Loop Filter (DLF). The modulator converts the fractional input into a sequence of integer control values whose long-term average accurately represents the desired fractional division ratio while significantly reducing the in-band quantization noise.

Fig.2.6 illustrate Comparison between the first-order Sigma-Delta Modulator and the MASH 1-1, the proposed MASH 1-1 architecture provides superior spectral performance by shaping the quantization noise with a second-order Noise Shaping. Consequently, a larger portion of the quantization noise is shifted outside the loop bandwidth, where it can be effectively attenuated by the low-pass characteristics of the DPLL. This improvement directly contributes to reduced phase noise, lower fractional spurs, and enhanced frequency synthesis accuracy.

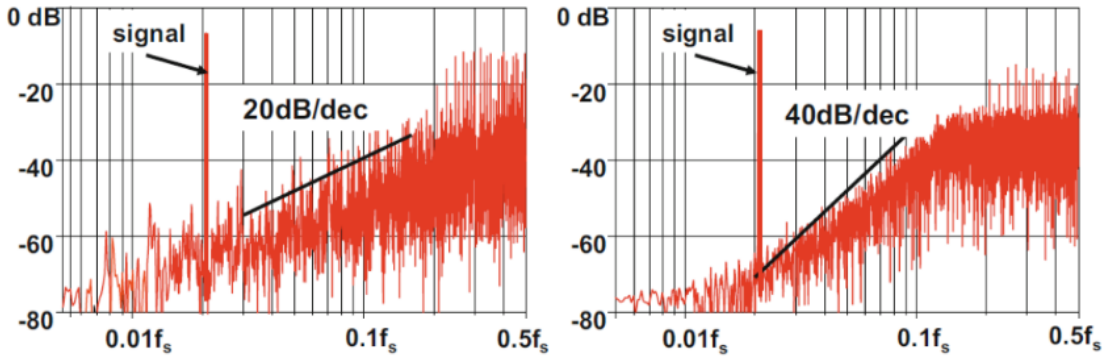


Fig 2.6: Comparison of the quantization noise spectrum for the First-Order DSM and the proposed MASH 1-1 architecture.

Fig.2.7 illustrates the simulation results confirmed the correct operation of both accumulator stages, the digital noise cancellation logic, and the expected second-order noise-shaping behavior.

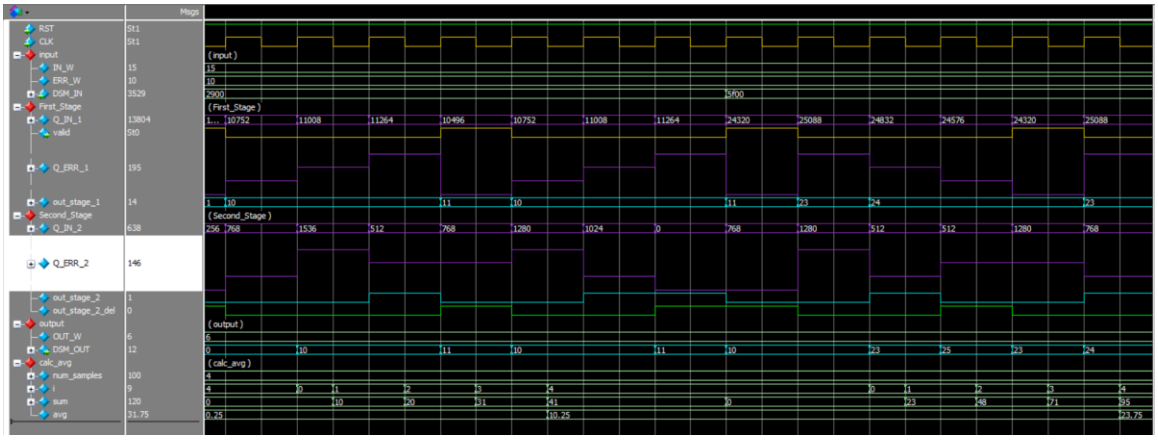


Fig2.7: Simulation waveform showing the output

2.5 LMS Correlator

2.5.1 Introduction

The Least Mean Squares (LMS) algorithm is one of the most widely used adaptive algorithms due to its simple implementation, low computational complexity, and efficient convergence characteristics. In digital integrated circuits, LMS-based correlators are frequently employed for adaptive calibration, where unknown system parameters are continuously estimated and updated to compensate for implementation imperfections.

In the proposed Fractional-N Digital Phase-Locked Loop (DPLL), the LMS correlator is employed as a digital calibration block within the divider subsystem. Instead of directly controlling the oscillator frequency, the LMS continuously estimates a digital calibration coefficient that compensates for timing errors introduced by the programmable divider and the Digital-to-Time Converter (DTC).

2.5.2 Operating Principle

Fig.2.8 illustrates the block diagram of LMS. In the divider path of the proposed DPLL, the First-Order Sigma-Delta Modulator generates a

quantization error (e_Q). Due to the periodic nature of the divider operation, this error can become correlated with the timing error observed by the Time-to-Digital Converter (TDC), resulting in increased fractional spurs and degradation of the synthesized clock quality.

To reduce this effect, the proposed LMS correlator continuously evaluates the correlation between the TDC output and the quantization error generated by the First-Order Sigma-Delta Modulator. Based on this correlation, the LMS adaptively updates a digital calibration coefficient (K_{CAL}), which represents the estimated correction required to compensate for the divider-induced timing error.

The generated calibration coefficient is subsequently multiplied by the instantaneous quantization error(e_Q) to produce a Digital Control Word (DCW). Rather than controlling the Digitally Controlled Oscillator (DCO), this DCW is applied to the Digital-to-Time Converter (DTC), where it adjusts the output delay of the feedback clock. By adaptively modifying the DTC delay, the proposed calibration loop compensates for timing mismatches introduced by the divider, thereby reducing fractional spurs and improving the overall timing accuracy of the DPLL.

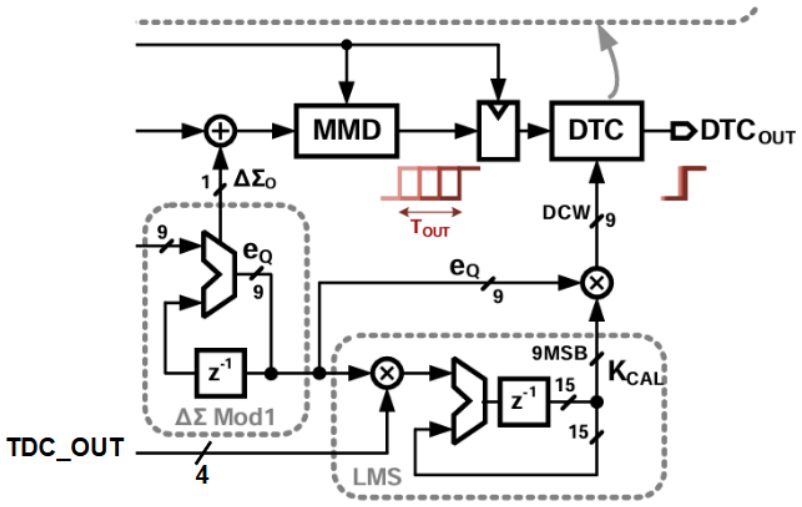


Fig.2.8: Proposed divider calibration subsystem showing the interaction between the First-Order DSM, LMS correlator, and DTC.

Fig.2.9 illustrates the simulation results confirmed the correct operation of accumulator stage, determine K_{CAL} successfully by implementing the LMS update equation:

$$K_CAL[n] = K_CAL[n-1] + \mu * Q_ERR[n-1] * TDC_OUT[n]$$

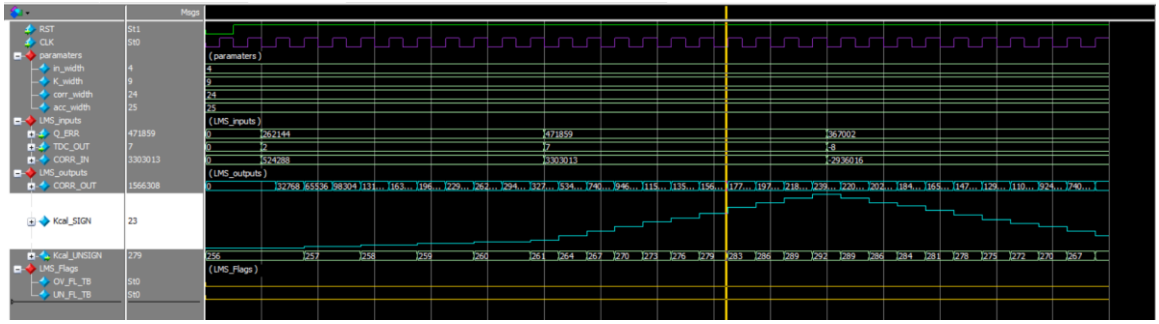


Fig2.9: Simulation waveform showing the output

Chapter 3

Integration and ASIC Implementation of the Digital Subsystems

3.1 Introduction

Following the individual design and functional verification of the digital building blocks presented in Chapter 2, the next stage of the proposed work involves integrating these modules into functional subsystems and evaluating their operation at the system level. System-level integration is a critical step in the design process, as it verifies the correct interaction between individual modules before proceeding to physical implementation.

Two digital subsystems were developed as part of the proposed Fractional-N Digital Phase-Locked Loop (DPLL). The first subsystem, referred to as DIG_FEEDFWD, integrates the Digital Loop Filter (DLF) with the Second-Order Sigma-Delta Modulator based on the MASH 1-1 architecture to generate the Digital Control Word (DCW) required by the Digitally Controlled Oscillator (DCO). The second subsystem, referred to as DIG_FEEDBACK, combines the LMS Correlator with the First-Order Sigma-Delta Modulator to generate the digital calibration signals used by the Digital-to-Time Converter (DTC).

Both subsystems were first verified through RTL simulation to ensure correct functionality and proper communication between their constituent modules. After successful functional verification, each subsystem was implemented using the complete ASIC design flow. The implementation included logic synthesis, static timing analysis (STA), physical implementation, and layout verification.

This chapter presents the integration methodology adopted for both digital subsystems and discusses the implementation results obtained throughout the complete RTL-to-GDSII ASIC design flow.

3.2 Integration of the DIG_FEEDFWD Subsystem

The DIG_FEEDFWD subsystem integrates the Digital Loop Filter (DLF) with the Second-Order Sigma-Delta Modulator based on the MASH 1-1 architecture. The DLF processes the digital phase error and generates the Digital Control Word (DCW), while the MASH modulator converts the fractional component of the DCW into a sequence of integer control values required by the Digitally Controlled Oscillator (DCO). This integrated architecture forms the digital feed-forward path of the proposed DPLL.

Prior to physical implementation, the integrated subsystem was verified using RTL simulation to ensure correct communication between the DLF and the MASH 1-1 modulator and to validate the generated control sequence under different operating conditions. The functional simulation waveform of the integrated subsystem is presented in Figure 3.1.

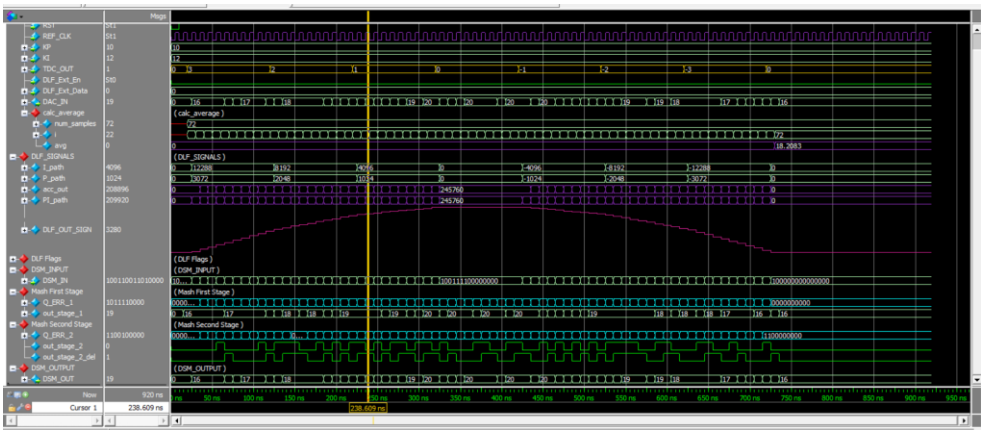


Fig3.1: Functional simulation waveform of the integrated DIG_FEEDFWD subsystem.

3.3 Integration of the DIG_FEEDBACK Subsystem

The second integrated subsystem, referred to as DIG_FEEDBACK, combines the LMS Correlator with the First-Order Sigma-Delta Modulator to realize the adaptive calibration path of the proposed DPLL.

Within this subsystem, the First-Order Sigma-Delta Modulator generates the quantization error, while the LMS correlator continuously estimates the calibration coefficient KCAL by evaluating the correlation between the quantization error and the output of the Time-to-Digital Converter (TDC). The estimated calibration coefficient is then multiplied by the instantaneous quantization error to generate the Digital Control Word (DCW), which controls the Digital-to-Time Converter (DTC) and compensates for divider-induced timing errors.

Before proceeding to the ASIC implementation flow, the integrated DIG_FEEDBACK subsystem was verified through RTL simulation to ensure the correct interaction between the LMS correlator and the First-Order Sigma-Delta Modulator. The simulation results confirmed the successful generation of the calibration coefficient, the Digital Control Word (DCW), and the associated control signals. The functional simulation waveform is shown in Figure 3.2.

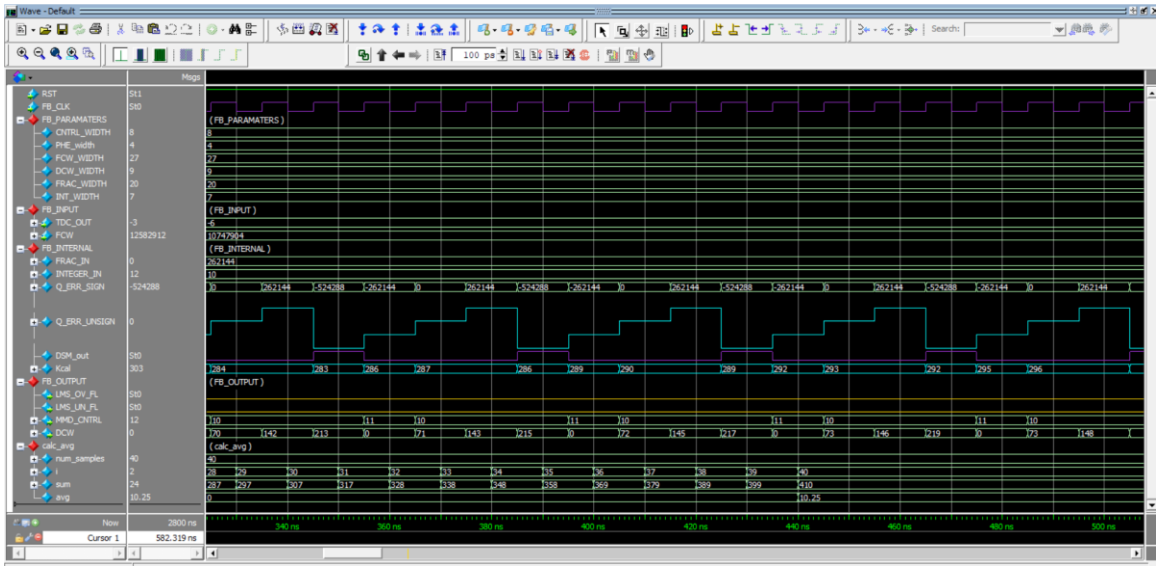


Figure 3.2: Functional simulation waveform of the integrated DIG_FEEDBCK subsystem.

3.4 Logic Synthesis

Logic synthesis represents the first stage of the ASIC implementation flow, where the verified RTL description is translated into a technology-dependent gate-level netlist using a standard-cell library. During this stage, the synthesis tool maps the behavioral Verilog description into logic gates while optimizing the implementation to satisfy the specified timing, area, and power constraints.

In this work, both integrated digital subsystems, DIG_FEEDFWD and DIG_FEEDBACK, were synthesized using the TSMC 130 nm CMOS standard-cell technology library. Design constraints were specified according to the target operating frequencies of each subsystem. The DIG_FEEDFWD subsystem was synthesized to operate at 50 MHz, whereas the DIG_FEEDBACK subsystem was synthesized with a target operating frequency of 100 MHz, reflecting the timing requirements of each subsystem within the proposed DPLL architecture.

Following synthesis, a technology-mapped gate-level netlist was successfully generated for each subsystem. Figures 3.3 and 3.4 present the synthesized gate-level schematics of the DIG_FEEDFWD and DIG_FEEDBACK top modules, respectively, as generated by Synopsys Design Compiler.

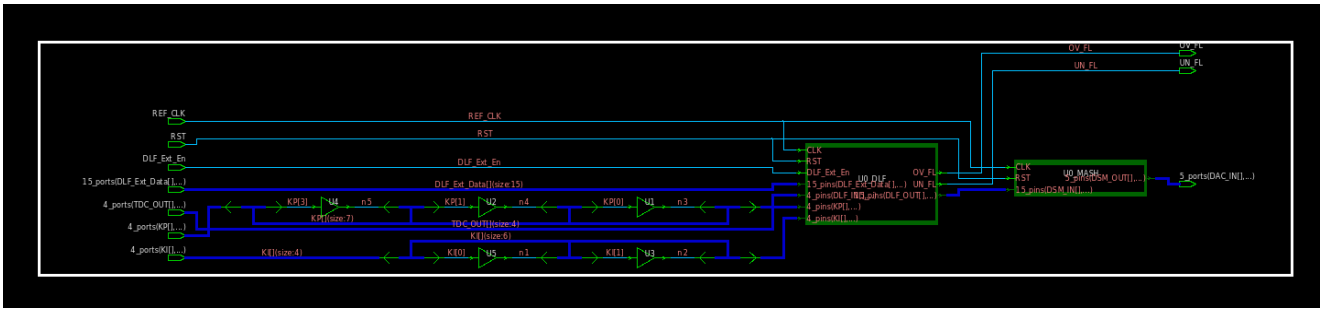


Figure 3.3: Technology-mapped gate-level schematic of the DIG_FEEDFWD subsystem generated by Synopsys Design Compiler.

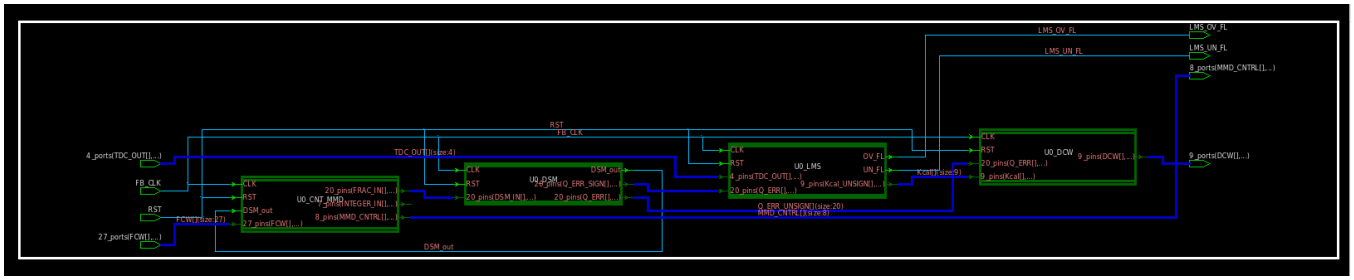


Figure 3.4: Technology-mapped gate-level schematic of the DIG_FEEDBACK subsystem generated by Synopsys Design Compiler.

After synthesis, several implementation reports were generated to evaluate the quality of the synthesized designs. These reports include the area report, which summarizes the silicon area occupied by each subsystem, and the power report, which provides estimates of both dynamic and leakage power consumption. These reports are presented and discussed in the following sections.

Furthermore, timing analysis performed during synthesis confirmed that both integrated subsystems successfully satisfied the specified timing constraints. No setup or hold timing violations were reported for any timing path, indicating that the synthesized designs were timing-clean and suitable for subsequent stages of the ASIC implementation flow.

3.4.1 Area Report

The synthesized designs were evaluated in terms of silicon area using the area report generated by Synopsys Design Compiler. The reported area values are expressed in square micrometers (μm^2), providing an estimate of the physical silicon occupied by each synthesized subsystem.

The DIG_FEEDFWD subsystem occupies a total cell area of 4964.497 μm^2, while the DIG_FEEDBACK subsystem occupies 12622.46 μm^2. The corresponding area reports are presented in Figures 3.5 and 3.6, respectively.

```
                                Global cell area          Local cell area
                                ------------------  ----------------------------
Hierarchical cell               Absolute   Percent  Combi-     Noncombi-  Black-
                                Total      Total    national   national   boxes   Design
-------------------------------  ---------  -------  ---------  ---------  ------  ----------------------------------------
DIG_FEEDFWD                     4964.4973    100.0    38.8311     0.0000  0.0000  DIG_FEEDFWD
U0_DLF                          3252.3988     65.5  2605.2138   647.1850  0.0000  DLF_in_width4_coeff_width4_out_width15
U0_MASH                         1673.2674     33.7   797.8026   875.4648  0.0000  DSM_MASH_1_1_IN_W15
-------------------------------  ---------  -------  ---------  ---------  ------  ----------------------------------------
Total                                                3441.8475  1522.6498  0.0000
```

Figure 3.5: Area report of the DIG_FEEDFWD subsystem.

```
                                Global cell area          Local cell area
                                ------------------  ----------------------------
Hierarchical cell               Absolute   Percent  Combi-     Noncombi-  Black-
                                Total      Total    national   national   boxes   Design
-------------------------------  ---------  -------  ---------  ---------  ------  ----------------------------------------
DIG_FEEDBCK                     12622.4609    100.0     0.0000     0.0000  0.0000  DIG_FEEDBCK
U0_CNT_MMD                       1073.1504      8.5    91.7826   981.3678  0.0000  CNTRL_MMD_FCW_WIDTH27_FRAC_WIDTH20_INT_WIDTH7_CNTRL_WIDTH8
U0_DCW                           5088.0508     40.3  4770.3419   317.7090  0.0000  CALC_DCW_DCW_WIDTH9_err_width20_K_width9
U0_DSM                           1374.3856     10.9   778.9754   595.4102  0.0000  DSM_1ST_ORDER_in_width20
U0_LMS                           5086.8741     40.3  4432.6289   654.2452  0.0000  LMS_in_width4_K_width9_err_width20
-------------------------------  ---------  -------  ---------  ---------  ------  ----------------------------------------
Total                                                10073.7288  2548.7322  0.0000
```

Figure 3.6: Area report of the DIG_FEEDBACK subsystem.

3.4.2 Power Report

Power consumption was estimated after logic synthesis using the power analysis report generated by Synopsys Design Compiler. The reported power values are expressed in milliwatts (mW) and include both dynamic and leakage power components which Dynamic Power Units = 1mW and Leakage Power Units = 1pW.

The estimated total power consumption of the DIG_FEEDFWD subsystem is 0.165 mW, whereas the DIG_FEEDBACK subsystem consumes 1.133mW under the specified operating conditions. The detailed power reports are shown in Figures 3.7 and 3.8, respectively.

```
--------------------------------------------------------------------------------
                                   Switch   Int      Leak     Total
Hierarchy                          Power    Power    Power    Power    %
--------------------------------------------------------------------------------
DIG_FEEDFWD                        8.06e-02 8.13e-02 3.31e+06    0.165 100.0
  U0_MASH (DSM_MASH_1_1_IN_W15)    3.17e-02 5.03e-02 1.17e+06 8.31e-02  50.3
  U0_DLF (DLF_in_width4_coeff_width4_out_width15)
                                   4.64e-02 3.07e-02 2.07e+06 7.92e-02  47.9
1
```

Figure 3.7: Power report of the DIG_FEEDFWD subsystem.

```
33
34 ------------------------------------------------------------------------------
35                                      Switch   Int      Leak     Total
36 Hierarchy                            Power    Power    Power    Power    %
37 ------------------------------------------------------------------------------
38 DIG_FEEDBCK                             0.719    0.404 9.83e+06    1.133 100.0
39   U0_LMS (LMS_in_width4_K_width9_err_width20)
40                                         0.283    0.135 4.17e+06    0.422  37.3
41   U0_CNT_MMD (CNTRL_MMD_FCW_WIDTH27_FRAC_WIDTH20_INT_WIDTH7_CNTRL_WIDTH8)
42                                         3.64e-02 9.15e-02 5.45e+05    0.129  11.3
43   U0_DCW (CALC_DCW_DCW_WIDTH9_err_width20_K_width9)
44                                         0.278    0.112 4.25e+06    0.395  34.9
45   U0_DSM (DSM_1ST_ORDER_in_width20)     0.121 6.57e-02 8.68e+05    0.187  16.5
46 1
```

Figure 3.8: Power report of the DIG_FEEDBACK subsystem.

The obtained results indicate the estimated power characteristics of each synthesized subsystem and provide an initial assessment of their implementation efficiency prior to physical design.

3.5 Formal Verification

Formal verification was performed to ensure the logical equivalence between the reference design and the implemented gate-level netlist after the implementation flow. Unlike simulation-based verification, formal verification exhaustively compares the functionality of both designs without relying on input test vectors, providing a mathematically complete proof of equivalence.

In this work, formal equivalence checking was carried out for both integrated subsystems, **DIG_FEEDFWD** and **DIG_FEEDBACK**, using **Synopsys Formality**. The verification process compared the synthesized gate-level implementation against the corresponding RTL descriptions to ensure that all logic transformations introduced during synthesis preserved the intended functionality.

The verification results confirmed that **no failing points** were detected in either subsystem. All compare points were successfully matched, demonstrating complete functional equivalence between the RTL and the synthesized gate-level netlists.

The formal verification failing points reports for the **DIG_FEEDFWD** and **DIG_FEEDBACK** subsystems are presented in **Figures 3.9** and **3.10**, respectively.

```
1 ************************************************
2 Report         : failing_points
3
4 Reference      : Ref:/WORK/DIG_FEEDFWD
5 Implementation : Imp:/WORK/DIG_FEEDFWD
6 Version        : L-2016.03-SP1
7 Date           : Mon Jul 13 04:55:36 2026
8 ************************************************
9
10 No failing compare points.
11
12 1
```

Figure 3.9: Formal verification failing points report for the DIG_FEEDFWD subsystem.

```
1 ************************************************
2 Report         : failing_points
3
4 Reference      : Ref:/WORK/DIG_FEEDBCK
5 Implementation : Imp:/WORK/DIG_FEEDBCK
6 Version        : L-2016.03-SP1
7 Date           : Mon Jul 13 05:05:32 2026
8 ************************************************
9
10 No failing compare points.
11
12 1
```

Figure 3.10: Formal verification failing points report for the DIG_FEEDBACK subsystem.

Furthermore, the verification reports indicate that all comparison points were classified as **passing points**, with **zero failing points** reported throughout the verification process. The detailed passing-point reports for both subsystems are shown in **Figures 3.11** and **3.12**, confirming that every registered output and internal comparison point satisfied the equivalence criteria.

```
218 ********************************* Verification Results *********************************
219 Verification SUCCEEDED
220    ATTENTION: synopsys_auto_setup mode was enabled.
221               See Synopsys Auto Setup Summary for details.
222 ---------------------------------------------------------
223  Reference design: Ref:/WORK/DIG_FEEDFWD
224  Implementation design: Imp:/WORK/DIG_FEEDFWD
225  64 Passing compare points
226 ------------------------------------------------------------------------------------------
227 Matched Compare Points     BBPin    Loop   BBNet     Cut    Port     DFF     LAT   TOTAL
228 ------------------------------------------------------------------------------------------
229 Passing (equivalent)           0       0       0       0       7      57       0      64
230 Failing (not equivalent)       0       0       0       0       0       0       0       0
231 ******************************************************************************************
```

Figure 3.11: Passing-point report for the DIG_FEEDFWD subsystem.

```
237 ********************************* Verification Results *********************************
238 Verification SUCCEEDED
239    ATTENTION: synopsys_auto_setup mode was enabled.
240               See Synopsys Auto Setup Summary for details.
241 ---------------------------------------------------------
242  Reference design: Ref:/WORK/DIG_FEEDBCK
243  Implementation design: Imp:/WORK/DIG_FEEDBCK
244  110 Passing compare points
245 ------------------------------------------------------------------------------------------
246 Matched Compare Points     BBPin    Loop   BBNet     Cut    Port     DFF     LAT   TOTAL
247 ------------------------------------------------------------------------------------------
248 Passing (equivalent)           0       0       0       0      19      91       0     110
249 Failing (not equivalent)       0       0       0       0       0       0       0       0
250 ******************************************************************************************
```

Figure 3.12: Passing-point report for the DIG_FEEDBACK subsystem.

The successful completion of formal verification provides strong confidence that the synthesized implementations faithfully preserve the functional behavior of their corresponding RTL descriptions and are therefore suitable for subsequent stages of the ASIC implementation flow.

3.6 Design for Testability (DFT) Implementation

To improve the testability of the proposed digital subsystems, Design for Testability (DFT) techniques were applied using scan-chain insertion. The DFT process was performed on both the DIG_FEEDFWD and DIG_FEEDBACK subsystems after logic synthesis. During scan insertion, a scan clock frequency of 50 MHz was specified for both subsystems, providing a common test operating condition throughout the DFT flow.

Following scan insertion, DFT Design Rule Checking (DFT DRC) was performed to verify the correctness of the inserted scan structures and to

ensure that all scan-related design rules were satisfied. The generated post-DFT reports confirmed that both subsystems successfully passed the DFT verification process without any reported violations.

```
17
18 ---------------------------------------------------------------
19   DRC Report
20
21   Total violations: 0
22
23 ---------------------------------------------------------------
24
25
26 Test Design rule checking did not find violations
27
28 ---------------------------------------------------------------
29   Sequential Cell Report
30
31   0 out of 57 sequential cells have violations
32
33 ---------------------------------------------------------------
34
```

Figure 3.13: Post-DFT DRC report show there is no violations for the DIG_FEEDFWD subsystem.

```
109      Uncollapsed Stuck Fault Summary Report
110  -----------------------------------------------
111  fault class                     code   #faults
112  ------------------------------  ----  ---------
113  Detected                         DT       2805
114  Possibly detected                PT          0
115  Undetectable                     UD          0
116  ATPG untestable                  AU         11
117  Not detected                     ND          0
118  -----------------------------------------------
119  total faults                              2816
120  test coverage                            99.61%
121  -----------------------------------------------
```

Figure 3.14: Post-DFT DRC report show the coverage percentage for the DIG_FEEDFWD subsystem.

```
18 ---------------------------------------------------------------
19   DRC Report
20
21   Total violations: 0
22
23 ---------------------------------------------------------------
24
25
26 Test Design rule checking did not find violations
27
28 ---------------------------------------------------------------
29   Sequential Cell Report
30
31   0 out of 110 sequential cells have violations
32
33 ---------------------------------------------------------------
```

Figure 3.15: Post-DFT DRC report show there is no violations for the DIG_FEEDBACK subsystem.

```
169
170      Uncollapsed Stuck Fault Summary Report
171  -----------------------------------------------
172  fault class                     code   #faults
173  ------------------------------  ----  ---------
174  Detected                         DT       7008
175  Possibly detected                PT          0
176  Undetectable                     UD          6
177  ATPG untestable                  AU         11
178  Not detected                     ND        155
179  -----------------------------------------------
180  total faults                              7180
181  test coverage                            97.69%
182  -----------------------------------------------
```

Figure 3.16: Post-DFT DRC report show the coverage percentage for the DIG_FEEDBACK subsystem.

3.6.1 Post-DFT Area Report

After scan insertion, the silicon area was re-evaluated to account for the additional scan logic introduced by the DFT process. The total post-DFT area of the DIG_FEEDFWD subsystem is 6404.77 μm², while the DIG_FEEDBACK subsystem occupies 17896.43 μm²

.

```
                                  Global cell area          Local cell area
                                  ------------------  ----------------------------
Hierarchical cell                 Absolute   Percent  Combi-     Noncombi-  Black-
                                  Total      Total    national   national   boxes   Design
-------------------------------- ---------  -------  ---------  ---------  ------  --------------------------------------------------
DIG_FEEDFWD                       6404.7782    100.0    35.3010     0.0000  0.0000  DIG_FEEDFWD
U0_DLF                            4266.7142     66.6  1573.2479   827.2201  0.0000  DLF_in_width4_coeff_width4_out_width15_test_1
U0_DLF/add_67                      620.1209      9.7   620.1209     0.0000  0.0000  DLF_in_width4_coeff_width4_out_width15_DW01_add_0
U0_DLF/add_77                     1246.1253     19.5  1246.1253     0.0000  0.0000  DLF_in_width4_coeff_width4_out_width15_DW01_add_1
U0_MASH                           2076.8755     32.4   354.1867  1055.4999  0.0000  DSM_MASH_1_1_IN_W15_test_1
U0_MASH/add_34                     374.1906      5.8   374.1906     0.0000  0.0000  DSM_MASH_1_1_IN_W15_DW01_add_1
U0_MASH/add_52                     292.9983      4.6   292.9983     0.0000  0.0000  DSM_MASH_1_1_IN_W15_DW01_add_0
U0_RST                              12.9437      0.2    12.9437     0.0000  0.0000  mux2X1_0
U0_clk                              12.9437      0.2    12.9437     0.0000  0.0000  mux2X1_1
-------------------------------- ---------  -------  ---------  ---------  ------  --------------------------------------------------
Total                                                  4522.0581  1882.7201  0.0000
```

Figure 3.17: Post-DFT area report of DIG_FEEDFWD.

```
                                  Global cell area           Local cell area
                                  ------------------  ----------------------------
Hierarchical cell                 Absolute    Percent  Combi-      Noncombi-  Black-
                                  Total       Total    national    national   boxes   Design
-------------------------------- ----------  -------  ----------  ---------  ------  ------------------------------------------------------------------
DIG_FEEDBCK                       17896.4304    100.0     56.4816     0.0000  0.0000  DIG_FEEDBCK
U0_CNT_MMD                         1521.4731      8.5    300.0585  1221.4146  0.0000  CNTRL_MMD_FCW_WIDTH27_FRAC_WIDTH20_INT_WIDTH7_CNTRL_WIDTH8_test_1
U0_DCW                             6568.3394     36.7     58.8350   378.8974  0.0000  CALC_DCW_DCW_WIDTH9_err_width20_K_width9_test_1
U0_DCW/mult_15                     6130.6070     34.3   5799.9543     0.0000  0.0000  CALC_DCW_DCW_WIDTH9_err_width20_K_width9_DW02_mult_0
U0_DCW/mult_15/FS_1                 330.6527      1.8    330.6527     0.0000  0.0000  CALC_DCW_DCW_WIDTH9_err_width20_K_width9_DW01_add_0
U0_DSM                             2160.4213     12.1    210.6293  1350.8517  0.0000  DSM_1ST_ORDER_in_width20_test_1
U0_DSM/add_29                       598.9403      3.3    598.9403     0.0000  0.0000  DSM_1ST_ORDER_in_width20_DW01_add_0
U0_LMS                             7565.0043     42.3    357.7168   827.2201  0.0000  LMS_in_width4_K_width9_err_width20_test_1
U0_LMS/add_51                      1094.3310      6.1   1094.3310     0.0000  0.0000  LMS_in_width4_K_width9_err_width20_DW01_add_0
U0_LMS/mult_38                     5285.7364     29.5   4670.3223     0.0000  0.0000  LMS_in_width4_K_width9_err_width20_DW02_mult_0
U0_LMS/mult_38/FS_1                 615.4141      3.4    615.4141     0.0000  0.0000  LMS_in_width4_K_width9_err_width20_DW01_add_1
U0_RST                               11.7670      0.1     11.7670     0.0000  0.0000  mux2X1_0
U0_clk                               12.9437      0.1     12.9437     0.0000  0.0000  mux2X1_1
-------------------------------- ----------  -------  ----------  ---------  ------  ------------------------------------------------------------------
Total                                                  14118.0466  3778.3838  0.0000
```

Figure 3.18: Post-DFT area report of DIG_FEEDBACK.

3.6.2 Post-DFT Power Report

Power analysis was repeated after scan insertion to evaluate the impact of the additional test circuitry. The estimated total power consumption of the DIG_FEEDFWD subsystem is 0.315 mW, whereas the DIG_FEEDBACK subsystem consumes 0.769 mW.

```
 ------------------------------------------------------------------------------------
                                      Switch   Int      Leak     Total
 Hierarchy                            Power    Power    Power    Power    %
 ------------------------------------------------------------------------------------
 DIG_FEEDFWD                             0.196    0.113 5.38e+06    0.315 100.0
   U0_MASH (DSM_MASH_1_1_IN_W15_test_1) 4.29e-02 5.27e-02 1.39e+06 9.70e-02  30.8
     add_34 (DSM_MASH_1_1_IN_W15_DW01_add_1)
                                      8.63e-03 5.65e-03 3.55e+05 1.46e-02   4.6
     add_52 (DSM_MASH_1_1_IN_W15_DW01_add_0)
                                      4.09e-03 4.46e-03 2.79e+05 8.83e-03   2.8
   U0_DLF (DLF_in_width4_coeff_width4_out_width15_test_1)
                                      7.77e-02 5.78e-02 3.90e+06    0.139  44.3
     add_77 (DLF_in_width4_coeff_width4_out_width15_DW01_add_1)
                                      5.56e-03 1.15e-02 1.63e+06 1.87e-02   6.0
     add_67 (DLF_in_width4_coeff_width4_out_width15_DW01_add_0)
                                      4.40e-03 5.30e-03 5.82e+05 1.03e-02   3.3
   U0_RST (mux2X1_0)                  1.18e-03 1.20e-04 1.34e+04 1.32e-03   0.4
   U0_clk (mux2X1_1)                  7.30e-02 1.81e-03 1.34e+04 7.49e-02  23.8
1
```

Figure 3.19: Post-DFT power report of DIG_FEEDFWD.

```
--------------------------------------------------------------------------------
                                    Switch   Int      Leak     Total
Hierarchy                           Power    Power    Power    Power    %
--------------------------------------------------------------------------------
DIG_FEEDBCK                            0.468    0.284 1.66e+07    0.769 100.0
  U0_LMS (LMS_in_width4_K_width9_err_width20_test_1)
                                       0.125    0.116 8.74e+06    0.250  32.5
    add_51 (LMS_in_width4_K_width9_err_width20_DW01_add_0)
                                    8.59e-03 1.42e-02 1.32e+06 2.41e-02   3.1
    mult_38 (LMS_in_width4_K_width9_err_width20_DW02_mult_0)
                                    9.49e-02 6.69e-02 6.74e+06    0.169  21.9
  U0_CNT_MMD (CNTRL_MMD_FCW_WIDTH27_FRAC_WIDTH20_INT_WIDTH7_CNTRL_WIDTH8_test_1)
                                    4.77e-02 5.59e-02 8.58e+05    0.104  13.6
  U0_DCW (CALC_DCW_DCW_WIDTH9_err_width20_K_width9_test_1)
                                       0.110 4.97e-02 5.61e+06    0.165  21.5
    mult_15 (CALC_DCW_DCW_WIDTH9_err_width20_K_width9_DW02_mult_0)
                                    7.61e-02 2.74e-02 5.31e+06    0.109  14.1
  U0_DSM (DSM_1ST_ORDER_in_width20_test_1)
                                    3.97e-02 5.90e-02 1.32e+06    0.100  13.0
    add_29 (DSM_1ST_ORDER_in_width20_DW01_add_0)
                                    8.73e-03 5.47e-03 5.72e+05 1.48e-02   1.9
  U0_RST (mux2X1_0)                 1.37e-04 9.66e-05 1.37e+04 2.47e-04   0.0
  U0_clk (mux2X1_1)                    0.142 2.52e-03 1.34e+04    0.144  18.7
```

Figure 3.20: Post-DFT power report of DIG_FEEDBACK.

3.6.3 Post-DFT Formal Verification

Formal equivalence checking was performed after scan insertion to verify that the added test logic did not alter the functional behavior of either subsystem. The verification results confirmed successful equivalence between the reference RTL and the post-DFT gate-level netlists.

No failing points were reported, and all comparison points were successfully classified as passing points for both DIG_FEEDFWD and DIG_FEEDBACK.

```
******************************** Verification Results ********************************
Verification SUCCEEDED
   ATTENTION: synopsys_auto_setup mode was enabled.
              See Synopsys Auto Setup Summary for details.
---------------------------------------------------------
 Reference design: Ref:/WORK/DIG_FEEDFWD
 Implementation design: Imp:/WORK/DIG_FEEDFWD
 64 Passing compare points
--------------------------------------------------------------------------------------
Matched Compare Points     BBPin    Loop   BBNet     Cut    Port     DFF     LAT   TOTAL
--------------------------------------------------------------------------------------
Passing (equivalent)           0       0       0       0       7      57       0      64
Failing (not equivalent)       0       0       0       0       0       0       0       0
Not Compared
  Don't verify                 0       0       0       0       1       0       0       1
**************************************************************************************
```

Figure 3.21: Formality passing-point report of the DIG_FEEDFWD subsystem.

```
255 ********************************* Verification Results *********************************
256 Verification SUCCEEDED
257    ATTENTION: synopsys_auto_setup mode was enabled.
258               See Synopsys Auto Setup Summary for details.
259 ---------------------------------------------------------
260  Reference design: Ref:/WORK/DIG_FEEDBCK
261  Implementation design: Imp:/WORK/DIG_FEEDBCK
262  129 Passing compare points
263 ---------------------------------------------------------------------------------------
264 Matched Compare Points     BBPin    Loop   BBNet     Cut    Port     DFF     LAT   TOTAL
265 ---------------------------------------------------------------------------------------
266 Passing (equivalent)           0       0       0       0      19     110       0     129
267 Failing (not equivalent)       0       0       0       0       0       0       0       0
268 ***************************************************************************************
```

Figure 3.22: Formality passing-point report of the DIG_FEEDBACK subsystem.

3.7 Physical Design Implementation (PnR)

3.7.1 Introduction

To demonstrate the complete physical implementation flow of the proposed digital design, the Place and Route **(**PnR**)** process was carried out on the **DIG_FEEDFWD** subsystem. This subsystem was selected as a representative implementation example since it contains the major digital control blocks of the proposed DPLL. The objective of this stage is to illustrate the complete ASIC physical design methodology rather than to repeat identical implementation steps for both integrated subsystems.

The physical implementation was performed using a fixed core area of **92 µm × 92 µm**, which was selected to provide sufficient utilization while accommodating the synthesized standard cells and routing resources.

3.7.2 Floorplanning

Floorplanning represents the first stage of the physical implementation process. During this stage, the overall chip dimensions, core area, input/output pin locations, and placement constraints were defined before cell placement. A proper floorplan is essential for achieving efficient routing, minimizing congestion, and improving timing closure throughout the subsequent implementation stages.

For the proposed DIG_FEEDFWD subsystem, a core area of 92 µm × 92 µm was adopted. The floorplan was generated after importing the synthesized gate-level netlist and the technology library into the physical design environment.

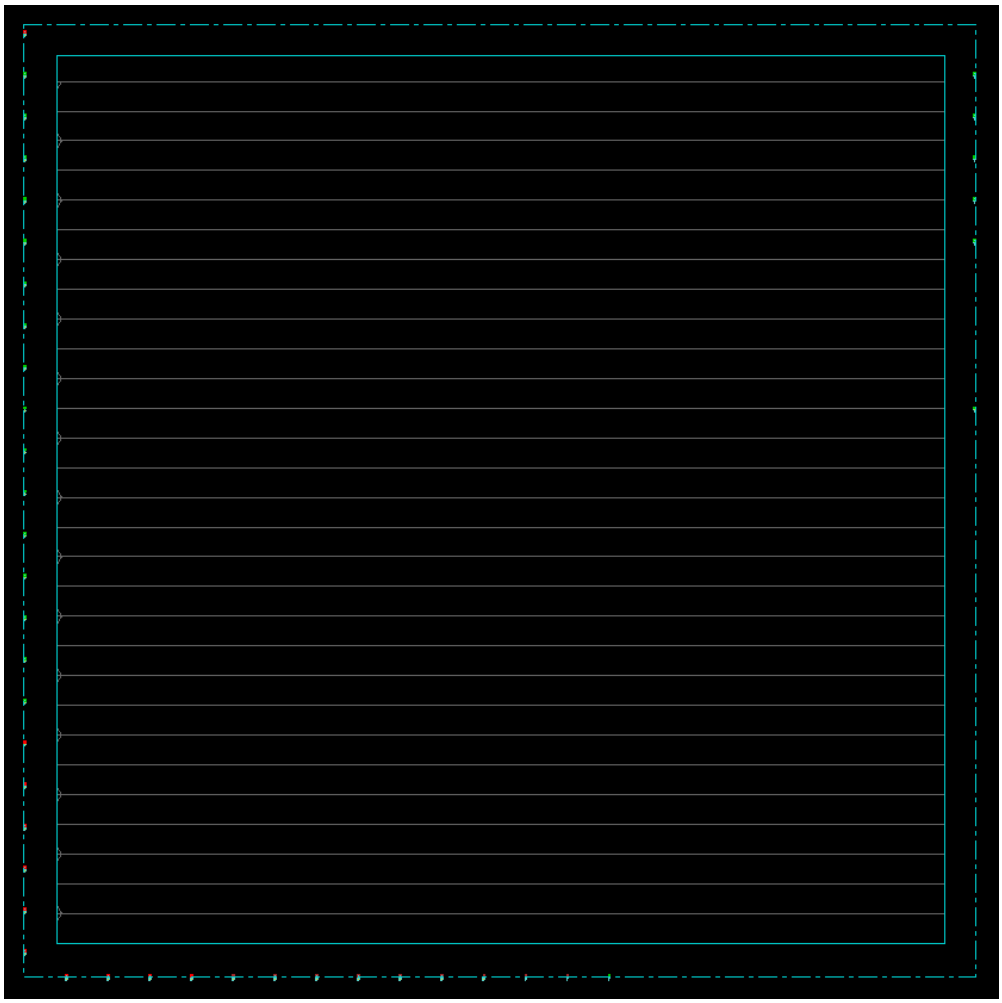


Figure 3.23: Floorplan of the DIG_FEEDFWD subsystem.

3.7.3 Power Planning

Following floorplanning, the power distribution network was established to ensure reliable delivery of the supply voltage to all standard cells. Power planning included the generation of the power rings and power stripes connected to the VDD and VSS networks, providing a low-resistance path

for current distribution while minimizing IR-drop and maintaining stable circuit operation.

The implemented power network for the **DIG_FEEDFWD** subsystem is shown in **Figure 3.24**.

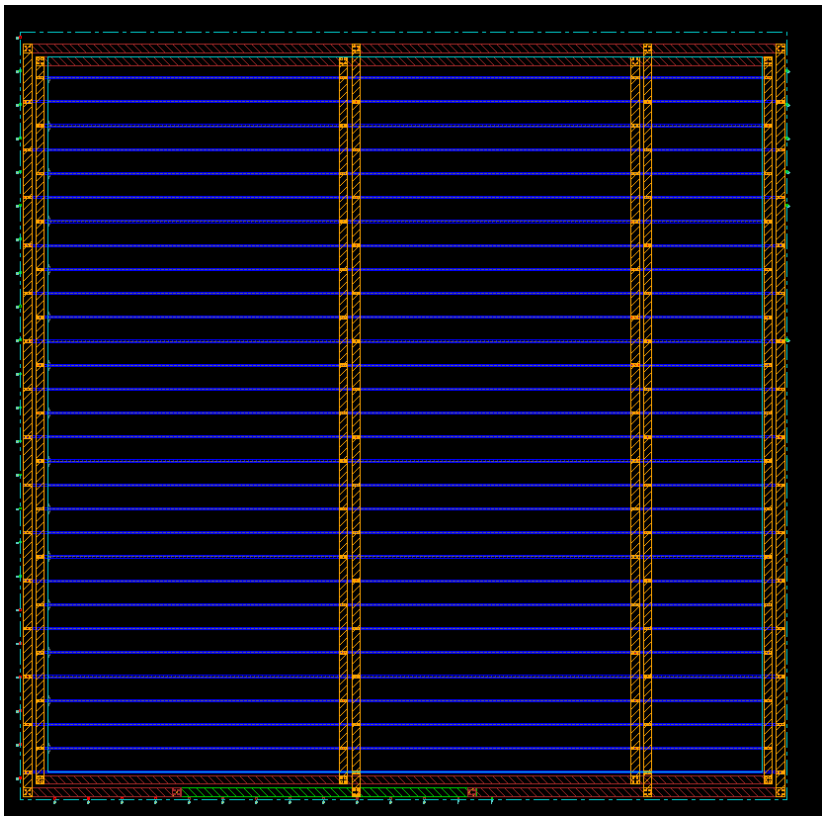


Figure 3.24: Power planning of the **DIG_FEEDFWD** subsystem.

3.7.4 Placement

During placement, all synthesized standard cells were positioned within the predefined floorplan while considering timing optimization, routing congestion, and area utilization. Placement optimization minimizes interconnect length and improves the timing characteristics before clock tree Synthesis (CTS).

The placement result of the **DIG_FEEDFWD** subsystem is presented in **Figure 3.25**.

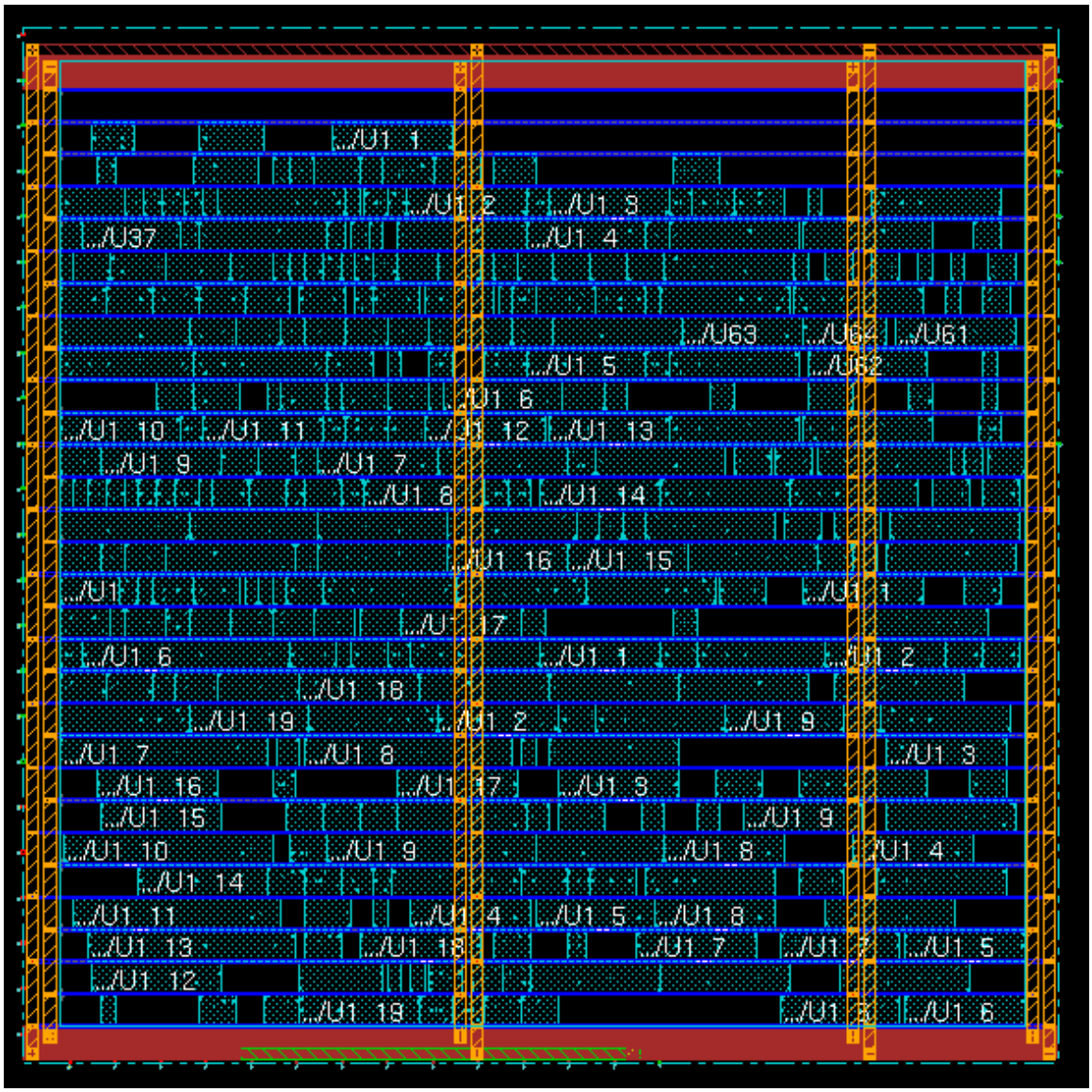


Figure 3.25: Placement result of the **DIG_FEEDFWD** subsystem.

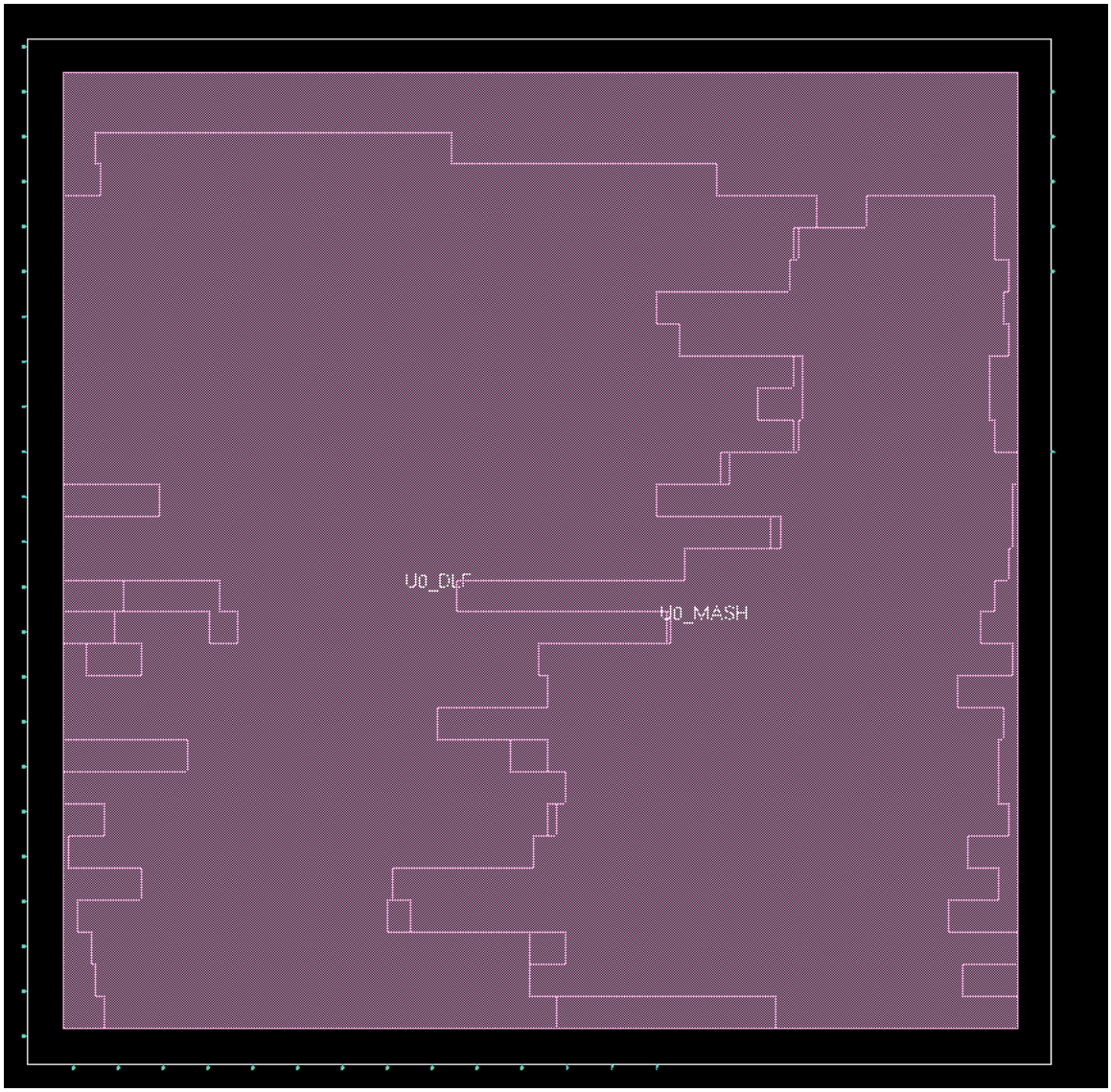


Figure 3.26: Amoeba view of the DIG_FEEDFWD subsystem.

3.7.5 Clock Tree Synthesis (CTS) & Routing

Clock Tree Synthesis (CTS) was performed to construct a balanced clock distribution network capable of delivering the clock signal to all sequential elements with controlled skew and insertion delay. Appropriate clock buffers were automatically inserted to satisfy the specified timing requirements and improve synchronization across the entire design.

Following CTS, signal routing was performed to establish all electrical interconnections between the placed standard cells using the available metal layers. The routing process was completed while satisfying timing constraints and technology design rules.

The final routed layout is shown in **Figure 3.27**.

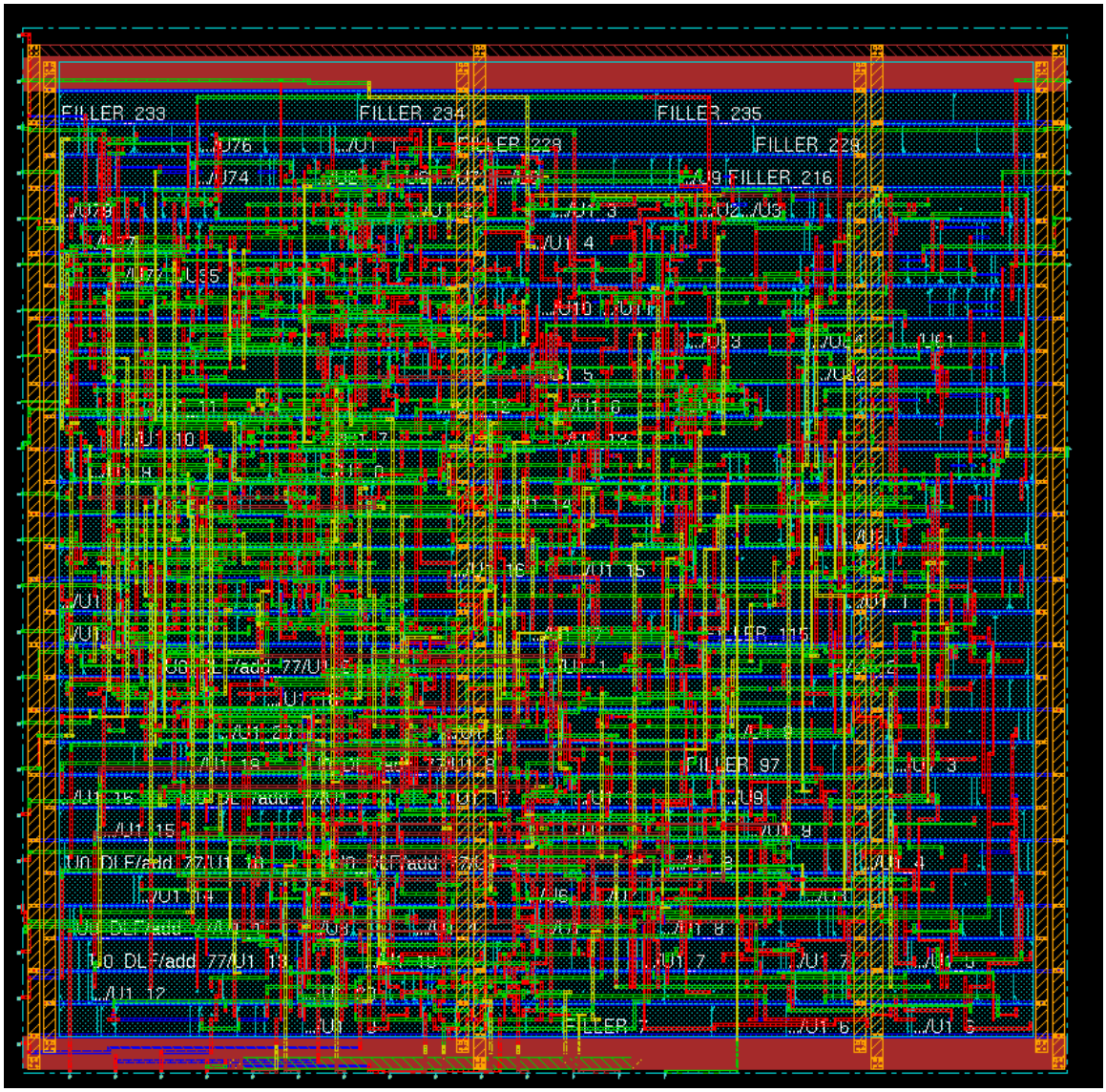


Figure 3.27: Final routed layout of the DIG_FEEDFWD subsystem.

3.7.6 Chip Finishing and Signoff

Following the routing stage, the physical design entered the chip finishing phase, during which the layout was prepared for final signoff verification and tape-out. This stage ensures that the implemented layout is physically complete, electrically correct, and free from manufacturing-related issues.

As part of the chip finishing process, filler cells were inserted to occupy the remaining gaps between adjacent standard cells. Although filler cells do not contribute to the functional behavior of the circuit, they play an essential role in maintaining the continuity of the N-well and power rails, satisfying fabrication requirements, and preventing DRC violations during manufacturing.

Prior to filler cell insertion, the layout was inspected to identify unused spaces between neighboring standard cells. Figure 3.25 illustrates the routed design before filler insertion, where the remaining gaps can be observed. After inserting the filler cells, all unused spaces were successfully eliminated, resulting in a continuous standard-cell row structure as shown in Figure 3.28.

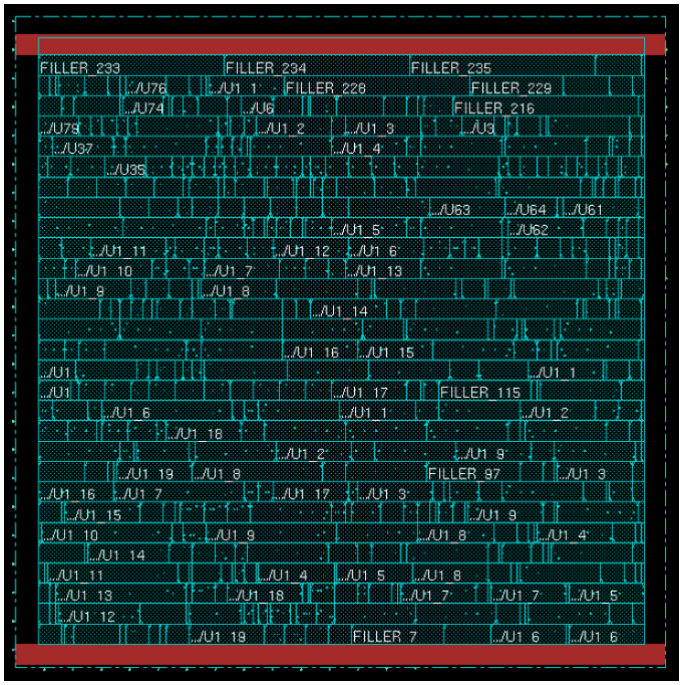


Figure 3.28: Routed layout after filler cell insertion demonstrating that all gaps have been successfully filled.

To further validate the physical implementation, several signoff verification checks were performed using the PnR tool. These checks ensure that the layout is electrically correct and satisfies the required timing and physical design constraints before GDSII generation.

The following verification steps were successfully completed:

- Connectivity Verification, confirming that all signal, power, and ground nets were correctly connected.
- Geometry Verification, ensuring compliance with the technology Design Rule Check (DRC) and identifying no geometry-related violations.
- Post-Route Setup Timing Analysis, verifying that all setup timing constraints were satisfied after routing.
- Post-Route Hold Timing Analysis, confirming that no hold timing violations were introduced during physical implementation.

The corresponding verification reports are presented in following figures.

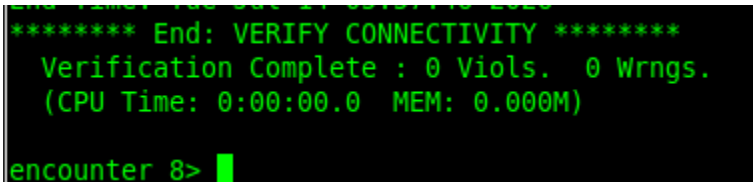


Figure 3.29: Connectivity verification successfully.

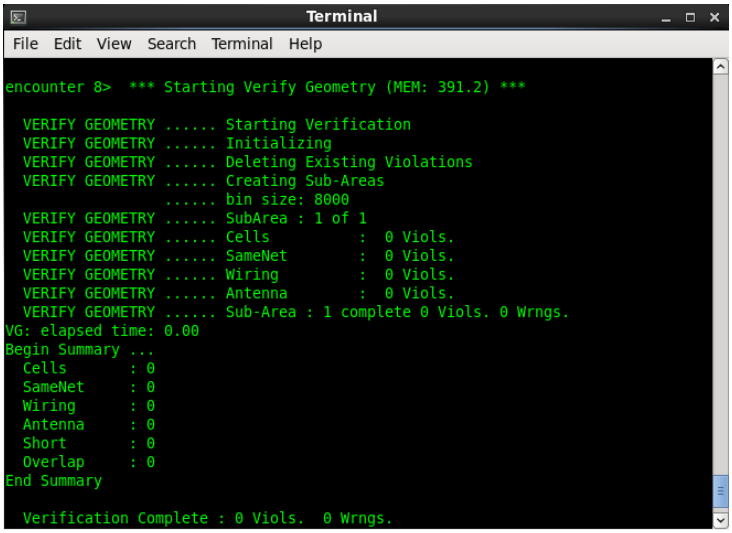


Figure 3.30: Geometry verification successfully

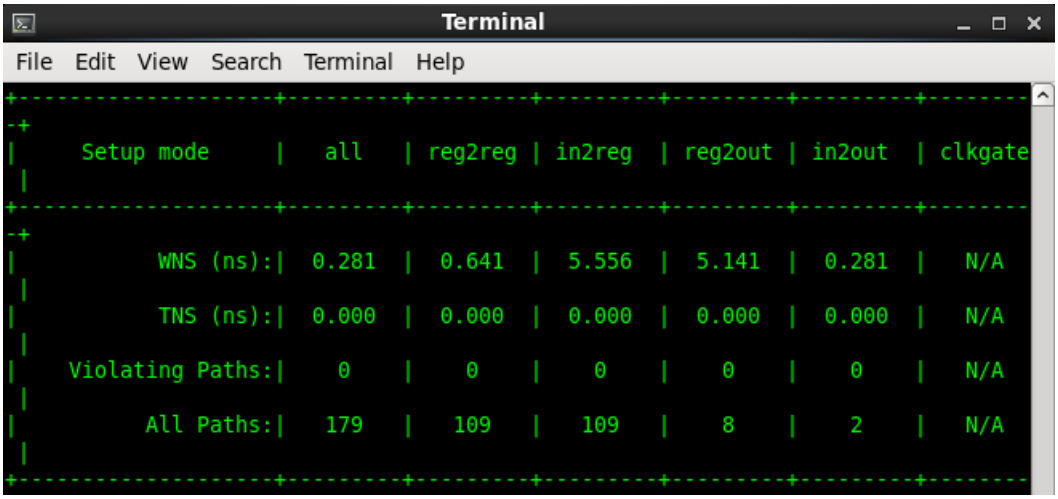


Figure 3.31: Post-route setup timing analysis confirming zero setup violations.

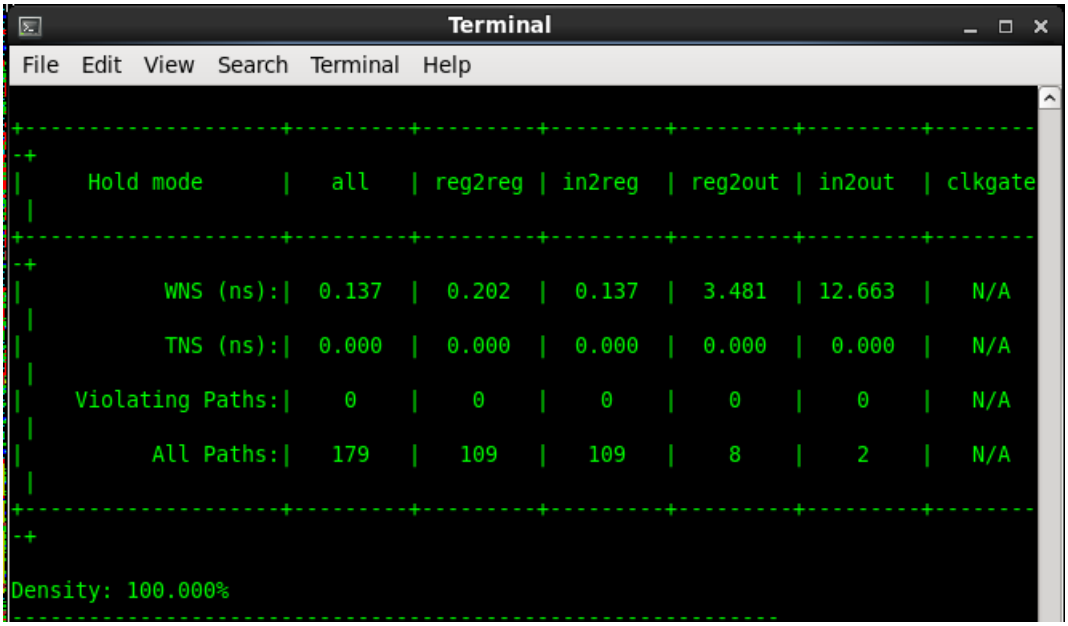


Figure 3.32: Post-route hold timing analysis confirming zero hold violations.

The successful completion of these signoff checks confirms that the physical implementation is free from connectivity, geometry, and timing violations, making the design ready for GDSII generation and fabrication.

3.7.7 Post-Layout Power Analysis

After completing the physical implementation, post-layout power analysis was performed to estimate the power consumption of the implemented design. The analysis accounts for the parasitic effects introduced during placement and routing (PnR), providing a more realistic estimation compared with synthesis-level power analysis.

The implemented DIG_FEEDFWD subsystem exhibits a total power consumption of 0.3387 mW, consisting of both dynamic and leakage power components. The detailed post-layout power report is presented in Fig 3.33.

Total Power

Total Internal Power:	0.1784	52.67%	
Total Switching Power:	0.1558	46.01%	
Total Leakage Power:	0.004469	1.319%	
Total Power:	0.3387		

Group	Internal Power	Switching Power	Leakage Power	Total Power	Percentage (%)
Sequential	0.1121	0.0147	0.0009269	0.1278	37.72
Macro	0	0	0	0	0
IO	0	0	0	0	0
Combinational	0.06625	0.1411	0.003542	0.2109	62.28
Total	0.1784	0.1558	0.004469	0.3387	100

Rail	Voltage	Internal Power	Switching Power	Leakage Power	Total Power	Percentage (%)
VDD	1.08	0.1784	0.1558	0.004469	0.3387	100

Clock	Internal Power	Switching Power	Leakage Power	Total Power	Percentage (%)
SCAN_CLK	0.01883	0.03431	0.0002623	0.05341	15.77
MASTER_CLK	0.02003	0.0208	0.0003024	0.04113	12.14
Total	0.03886	0.05511	0.0005647	0.09454	27.91

Figure 3.33: Post-layout power report of the DIG_FEEDFWD subsystem.

3.7.8 GDSII Release

Following successful signoff verification, the completed physical layout was exported in the **GDSII** format, which represents the final database delivered for semiconductor fabrication. The generated GDSII file contains the complete geometric description of the integrated circuit and serves as the final output of the ASIC physical design flow.

3.7.9 Conclusion

The successful completion of the physical implementation demonstrates that the proposed DIG_FEEDFWD subsystem is suitable for ASIC realization using the TSMC 130 nm technology. The design successfully completed all implementation stages, from floorplanning to GDSII generation, while satisfying the specified timing constraints and achieving the desired power and area objectives. This validates the practicality of the proposed digital architecture for integration within the complete DPLL system.